Dynamic Loading Assessment at the Fifth Metatarsal in Elite Athletes with a history of Jones Fracture

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Objectives: Fractures of the proximal metaphysis of the fifth metatarsal (i.e., Jones fractures) are common in the athletic population and frequently require surgical fixation to optimize healing rates and return to sport. Subtle hindfoot varus alignment, which likely results in greater forces at the fifth metatarsal, has been associated with increased risk of Jones fractures due to presumed increase in forces on the lateral foot. The goal of the present study was to dynamically assess plantar pressure loads at the fifth metatarsal in elite athletes with a history of Jones fracture. We hypothesize that athletes who have suffered Jones fractures will apply significantly higher loads at the fifth metatarsal base during athletic activities compared to matched uninjured athletes.
Methods: Sixteen elite, competitive athletes were recruited to participate. Eight athletes had a history of Jones fracture and eight age, gender and position-matched athletes without a history of foot injury were recruited as controls. While wearing calibrated, wireless pressure mapping insoles, all athletes performed a standardized series of athletic movements including walking, running and cutting. Tests were repeated with the patient's standard corrective insole. Peak pressure, mean pressure, maximum force, and force-time integral (i.e., impulse) were recorded for each activity. Comparison between groups was performed using a paired Wilcoxon Sign Rank test.
Results: Athletes with a history of Jones fracture showed a significantly increased peak pressure (183 ±23 kPA vs. 138±7 kPA), mean pressure (124±14 kPA vs. 95±4 kPA), and maximum force (15±1.2%BW vs. 12±1.2%BW) at the fifth metatarsal base during walking and running compared to uninjured matched controls (all $P<0.05$). Use of the corrective orthotic insole did not significantly reduce peak or mean pressure, maximum force, or impulse at the fifth metatarsal base during running or cutting.
Conclusion: Athletes with a history of Jones fracture exert significantly increased peak and mean forces at the base of the fifth metatarsal during common athletic activities. Standard corrective orthoses do not appear to offload this region in all cases. These increased loads may contribute to the development of stress injury to the fifth metatarsal during repetitive loading, and ultimately fracture of the bone. Identifying patients at risk for Jones fracture using dynamic loading assessment may facilitate an injury prevention program. Further study of orthotic devices, training programs, shoe wear, and devices designed to offload the fifth metatarsal is necessary.

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